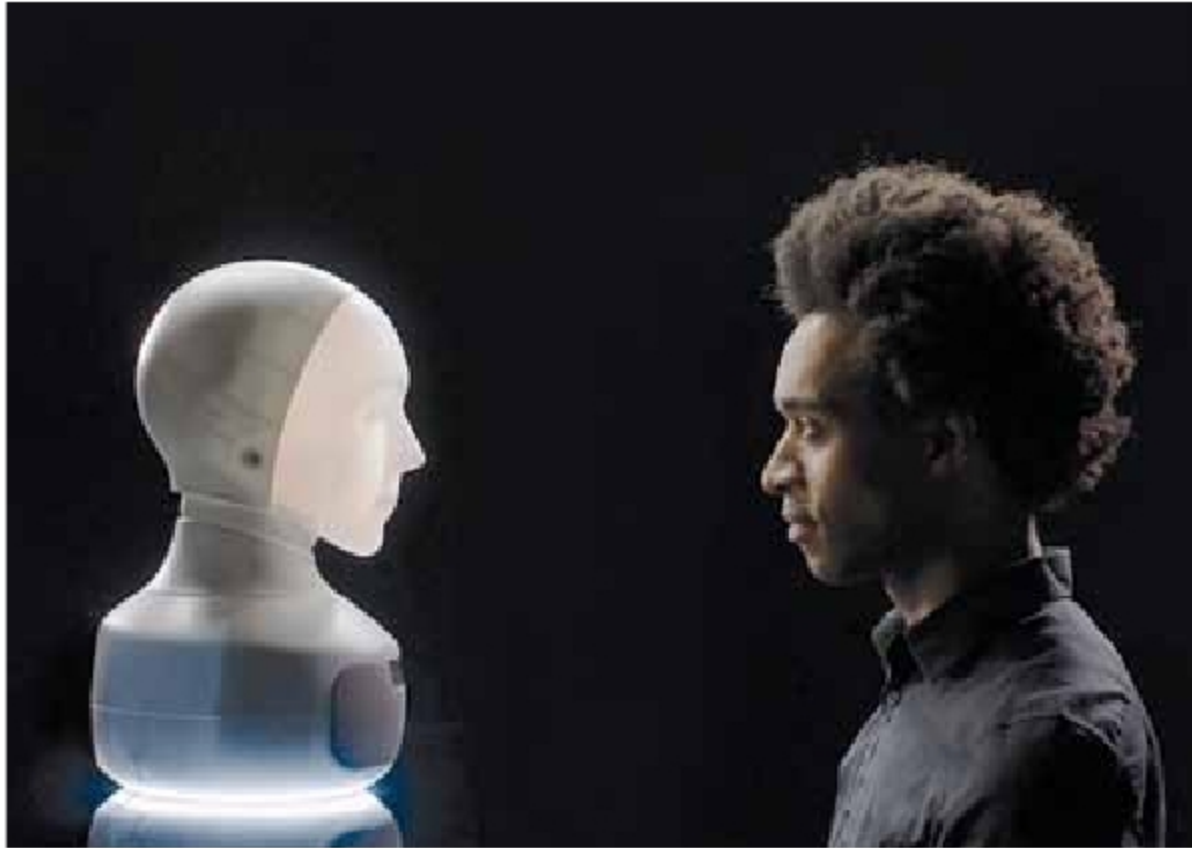




## An Amalgamation of Robotics and AI

Not knowing the local language can be a handicap for people trying to find the closest airport or railway station. This article talks about a social robot designed to solve this challenge.

**M**any a time, people face a language barrier when they go to an airport or railway station and try to find their transport. Can there be a different person for each language to guide people, or is there another way? Training employees of different origins, academics and grasping power is difficult for a human trainer. Such problems can be solved with a social robot.

Founded in 2014, Furhat Robotics, Sweden, has designed a social robot with conversational artificial intelligence (AI). The company aims to revolutionise human-computer interaction with conversational robots that can help humans interact with machines the way they do with each other. This robot can be useful in various

industries. It is capable of conducting competency-based interviews, which can help recruiting firms to hire candidates with the right skills regardless of their origin, colour, gender or nationality.

The robot is also capable of guiding people in an airport or a railway station in 40 different languages. This can reduce the human requirement for support services.

Organisations can also upskill their employees with the robot, as it is capable of providing training in varied areas, too.

This futuristic robot has a human-like face and makes subtle gestures. It can be programmed to create unique characters and personalities. One Swedish hiring firm, TNG, is set to use this robot to hire people. The robot will have intuitive

interactions with interviewees and then convert the interactions into transcripts, which will go to decision makers. This will ensure the decision makers will have only the questions and answers, and their decisions will not be influenced by gender, ethnicity and voice of the interviewees. The robot also holds the capability of interaction with multiple users.

Furhat conversational engine is designed on an advanced computational model of hierarchical state machines. It is capable of handling very-expressive and low-latency interactions. Furhat includes powerful tools to create faces and gestures; it can capture expressive movements from users and provide curative responses, too. It uses high-performance computer vision models for face recognition, and includes microphones and a camera for capturing voices and gestures. Furhat utilises projection beam technology on an opaque mask to create faces.

As of now, the company has discovered utility of the robot for recruiting, training and support services. The robot can specifically train employees on how to turn down a difficult client or refuse a loan. It has been deployed in such organisations as KPMG for dispensing financial advice, and has teamed up with Deutsche Bahn (railway company) for manning the desk at Germany's Frankfurt airport.

There are numerous other applications for Furhat. Such robots give an insight into the enormous potential of AI in robotics. **END** 🐧

**By: Deeksha Sharma**

The author was working as a technology journalist at EFY until recently.

*This article was originally published in the May 2019 issue of Electronics For You.*